

Titanic Dataset – Descriptive Statistics Task Report

Tools Used: Microsoft Excel and Google Colab (Python/Pandas)

Dataset: Titanic dataset with 891 passenger records. The analysis focuses on Age, Fare, SibSp and Parch.

1. Objective

To perform descriptive statistical analysis on selected numerical variables from the Titanic dataset using Excel and Python/Pandas.

2. Work Completed

- Loaded and verified the Titanic CSV dataset in Google Colab.
- Calculated mean, median, mode and standard deviation.
- Calculated the 25th, 50th and 75th percentiles.
- Calculated the Interquartile Range (IQR).
- Organized the results in Excel.

3. Descriptive Statistics Results

Variable	Mean	Median	Mode	Std Deviation	25th Percentile	50th Percentile	75th Percentile
Age	29.70	28.00	24.00	14.53	20.13	28.00	38.00
Fare	32.20	14.45	8.05	49.69	7.91	14.45	31.00
SibSp	0.52	0.00	0.00	1.10	0.00	0.00	1.00
Parch	0.38	0.00	0.00	0.81	0.00	0.00	0.00

4. Quartiles and IQR

Variable	Q1 (25th)	Q3 (75th)	IQR (Q3 – Q1)
Age	20.13	38.00	17.88
Fare	7.91	31.00	23.09
SibSp	0.00	1.00	1.00
Parch	0.00	0.00	0.00

5. Key Observations

- Average age is 29.70 years and the median age is 28.00 years.
- Fare has a mean of 32.20 and a median of 14.45.
- SibSp and Parch both have a median and mode of 0.00.
- Fare has the highest standard deviation among the selected variables.
- IQR represents the spread of the middle 50% of observations.

6. Conclusion

The Titanic dataset was successfully analyzed using Excel and Python/Pandas. The results summarize central tendency, variation and quartile distribution for Age, Fare, SibSp and Parch.

Formatting note: All numerical values in this report are displayed to exactly two decimal places to match the Excel sheet.